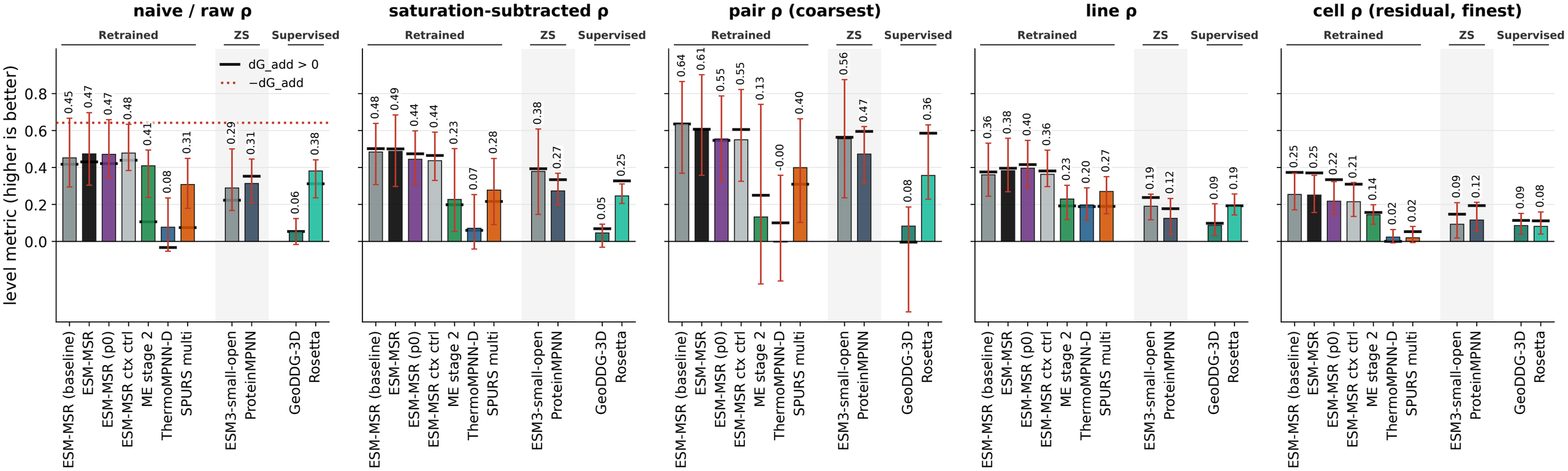

Epistasis = double-mutant prediction minus additive singles: ESM-MSR = 1/2(MT_AB+WT_AB) − WT_A − WT_B (no linking), ESM-MSR ctx ctrl = 1/2(~WT_AB+WT_AB) − WT_A − WT_B; ME stage 2, ThermoMPNN-D, SPURS multi minus the same seed's singles (3-seed means); ESM3-small-open unmasked 1/2(MT_AB+WT_AB) − WT_A − WT_B; ProteinMPNN, GeoDDG-3D, Rosetta minus sum of singles. Intervals: 95% bootstrap over domains.
Horizontal line behind each bar = the same level metric restricted to doubles with additive expectation dG_additive = dG_wt + ddG_A + ddG_B > 0 kcal/mol (7,405 test, 5,500 val doubles).
Sequential residual hierarchy (all 5 levels pooled; r = dddG − E[dddG | x] and r' = dddG' − E[dddG' | x'] via 25-quantile-bin smoothers): (1) naive / raw ρ: pooled over all doubles on raw dddG;
(2) saturation-subtracted ρ: pooled over all doubles on residuals r', r; (3) pair ρ: across position-pair matrices (≥20 doubles) of each pair's mean residual;
(4) line ρ: pooled over all row/column lines (≥3 cells) of pair-centred line means; (5) cell ρ (residual): pooled across all position-pair complete blocks (≥3×3) of global+pair+line-corrected residuals (c = r − r_row − r_col + r_pair) correlated by Spearman ρ.